

A Language Model Predicts the Next Token

A language model assigns probabilities to possible token sequences.



CORE CONCEPT

Tokenization Converts Text into IDs

A tokenizer converts text into tokens and integer token IDs from a fixed vocabulary. A token may be: Tokenization depends on the model. Character count and word count do not equal token count.

a complete word;

part of a word;

punctuation;

whitespace pattern;

special control marker.

RUN IT AND INSPECT THE RESULT

Embeddings Turn Token IDs into Vectors

```
input_ids = torch.tensor([[10, 20, 30]])  
embeddings = model.get_input_embeddings()(input_ids)  
print(embeddings.shape)
```

OUTPUT

```
torch.Size([1, 3, hidden_size])
```

There is one four-number vector per token.

REASONING SEQUENCE

Self-Attention Mixes Information Across Tokens

Frame

forms a query
describing what
information it seeks;

1

Transform

compares that query
with keys from
allowed tokens;

2

Validate

converts
comparisons into
attention weights;

3

Explain

combines value
vectors using those
weights.

4

Sentence: The bank raised its rate because it expected inflation. When representing the second it, attention may assign more weight to bank and surrounding context. The result is a context-dependent representation.

CORE CONCEPT

The Context Window Is a Finite Token Budget

The context window is the maximum token sequence the model can process in one request, including some combination of: Longer context consumes memory, latency, and cost. Content may be truncated or rejected when the limit is exceeded.

system and user messages;

conversation history;

retrieved documents;

tool results;

generated output.

RUN IT AND INSPECT THE RESULT

Logits Become Probabilities through Softmax

```
logits = torch.tensor([2.0, 1.0, 0.0])  
print(torch.softmax(logits, dim=0))
```

OUTPUT

```
tensor([0.6652, 0.2447, 0.0900])
```

$[7.39/11.11, 2.72/11.11, 1/11.11] \approx [0.665, 0.245, 0.090]$

CORE CONCEPT

Sampling Controls How a Token Is Chosen

Greedy decoding selects the highest-probability token each step. Sampling randomly selects according to a probability distribution. Temperature rescales logits before softmax:

lower T sharpens the distribution;

higher T flattens it;

temperature does not add knowledge or verify facts.

REASONING SEQUENCE

Base and Instruction Models Have Different Training Goals

Frame

A base model is pretrained mainly to continue token sequences.

1

Transform

Instruction models usually expect a model-specific chat template containing role and control tokens.

2

Validate

Sending plain text to an instruction model or chat-formatted text to a base model changes behaviour.

3

Explain

Neither model type guarantees truth, safety, or task correctness.

4

Prompt to a base model: Question: What is 2+2? Answer: may continue a learned document pattern. An instruction model receives structured messages such as user request and assistant turn, then generates the assistant response.

RUN IT AND INSPECT THE RESULT

Hallucination Is Unsupported Generation

```
if not supporting_chunks:  
    return {"answer": None, "reason": "insufficient_evidence"}
```

OUTPUT

A machine-readable refusal instead of an unsupported factual answer.

Unsafe answer invents a section number. Grounded answer states that the evidence does not contain the requested policy and cites what was checked.

QUALITY GATE

Quantization Trades Precision for Resource Use



lower memory use;



faster inference on supported hardware;



smaller model files;



some quality loss;

QUALITY GATE

Ollama Provides a Local Model Runtime



model identifier and version determine the artifact;



local runtime still consumes CPU, memory, and storage;



prompts and responses remain on the machine unless the application sends data elsewhere;



local does not automatically mean secure;

GUIDED LAB

Guided Lab: Trace One Generated Token

01

**tokenize a short
prompt and inspect
IDs**

02

**report input
token count**

03

**inspect model and
tokenizer identifiers**

04

**obtain
next-token
logits**